

Flutter Notes of 1st and 2nd Lecture

1. What is Flutter?

Flutter is a **free tool made by Google** used to **create mobile, web, and desktop apps** using a **single code**.

You write code once, and the app can run on **Android and iOS**.

In simple words:

Flutter helps developers make apps faster with one codebase.

2. History of Flutter

- Flutter was created by **Google**.
 - First introduced in **2017**.
 - Stable version released in **2018**.
 - It uses the **Dart programming language**.
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3. Why Use Flutter? (Simple Reasons)

1. **Single Code**
 - One code works for Android and iOS.
 2. **Fast Development**
 - Hot Reload lets you see changes instantly.
 3. **Beautiful UI**
 - Easy to design attractive screens.
 4. **Free and Open Source**
 - Anyone can use it without cost.
 5. **High Performance**
 - Apps run smoothly like native apps.
-

4. Where Flutter is Used

- Mobile apps
 - Web apps
 - Desktop apps
 - UI prototypes
-

5. Advantages of Flutter

- Saves time
- Easy UI design
- Large community
- Good documentation

Step to Create Flutter Project

1. Open **Visual Studio Code**
→ Click **File > New Window**
2. Go to **View**
3. Select **Command Palette**
4. Click **Flutter: New Project**
5. Select **Application**
6. Select the folder where you want to store the project
7. Enter the project name
8. Use **lowercase letters only**
9. Example: `my_first_app`
10. Wait until the project is created successfully

1 Stateless Widget & Stateful Widget

➤ Stateless Widget

Definition

A **StatelessWidget** is a widget that **does not change its state** during runtime. Once it is built, its UI remains the same.

✓ Used when:

- UI is fixed
- No data change
- No user interaction affecting UI

Key Points

- Immutable (cannot change)
- build() method is called once
- Faster performance

Syntax

```
class MyWidget extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return Widget;  
  }  
}
```

➤ Stateful Widget

Definition

A **StatefulWidget** is a widget that **can change its UI during runtime**.

✓ Used when:

- UI updates dynamically
 - Button clicks
 - Counters, forms, animations
-

Key Points

- Mutable
- Uses setState()
- Has two classes:
 - Widget class
 - State class

Syntax

```
class MyWidget extends StatefulWidget {  
  @override  
  _MyWidgetState createState() => _MyWidgetState();  
}  
  
class _MyWidgetState extends State<MyWidget> {  
  @override  
  Widget build(BuildContext context) {  
    return Widget;  
  }  
}
```

Example 1: Counter

```
class Counter extends StatefulWidget {  
  @override  
  _CounterState createState() => _CounterState();  
}  
  
class _CounterState extends State<Counter> {  
  int count = 0;  
  
  @override  
  Widget build(BuildContext context) {  
    return Column(  
      children: [  
        Text("Count: $count"),  
        ElevatedButton(  
          onPressed: () {  
            setState() {  
              count++;  
            }  
          }  
        ),  
        child: Text("Increment"),  
      ],  
    );  
  }  
}
```

2 Scaffold

Definition

Scaffold provides the **basic layout structure** of a Flutter app.

It supports:

- AppBar
- Body
- Drawer

- FloatingActionButton
- BottomNavigationBar
-

Syntax

```
Scaffold(  
  appBar: AppBar(),  
  body: Widget,  
)
```

Example 1: Simple Scaffold

```
Scaffold(  
  appBar: AppBar(title: Text("Home")),  
  body: Center(child: Text("Welcome")),  
);
```

3 AppBar

Definition

AppBar is the **top bar** of the app.

Used for:

- Title
- Actions
- Navigation

Syntax

```
AppBar(  
  title: Text("Title"),  
)
```

Example 1: Simple AppBar

```
AppBar(  
  title: Text("Flutter App"),  
);
```

Example 2: AppBar with Actions

```
AppBar(  
  title: Text("Home"),  
  actions: [  
    IconButton(  
      icon: Icon(Icons.search),
```

```
        onPressed: () {},  
      ),  
    ],  
  );
```

Example 3: Colored AppBar

```
AppBar(  
  title: Text("Custom"),  
  backgroundColor: Colors.green,  
);
```

4 Text Widget

Definition

Displays **text on the screen**.

Syntax

```
Text("Hello")
```

Example 1: Simple Text

```
Text("Flutter")
```

Example 2: Styled Text

```
Text(  
  "Flutter UI",  
  style: TextStyle(  
    fontSize: 20,  
    fontWeight: FontWeight.bold,  
  ),  
);
```

Example 3: Centered Colored Text

```
Text(  
  "Welcome",  
  style: TextStyle(color: Colors.blue),  
  textAlign: TextAlign.center,  
);
```

5 Color

Definition

Colors define the **visual appearance**.

Types of Colors

1. Material Colors

```
Colors.red  
Colors.blue
```

2. Custom Color

```
Color(0xFF00FF00)
```

3. Opacity Color

```
Colors.black.withOpacity(0.5)
```

Example 1

```
Container(color: Colors.orange)
```

Example 2

```
Text("Text", style: TextStyle(color: Color(0xFF2196F3)))
```

6 Center

Definition

Centers its child **both horizontally and vertically**.

Syntax

```
Center(child: Widget)
```

Example 1

```
Center(child: Text("Hello"))
```

Example 2

```
Center(  
  child: Container(  
    height: 100,  
    width: 100,  
    color: Colors.red,  
  ),  
);
```

7 SizedBox

Definition

Adds **fixed space or size**.

Syntax

```
SizedBox(height: 20)
```

Example 1: Space

```
SizedBox(height: 20)
```

Example 2: Sized Widget

```
SizedBox(  
  height: 100,  
  width: 100,  
  child: Container(color: Colors.blue),  
);
```

Example 3: Horizontal Space

```
SizedBox(width: 15)
```

Example 4: Simple Space Between Widgets

```
Column(  
  children: [  
    Text("First Line"),  
    SizedBox(height: 30), // Adds 30px vertical space  
    Text("Second Line"),  
  ],  
);
```

8 Container

Definition

A **box model widget** used for layout, decoration, margin, padding.

Syntax

```
Container(  
  height: 100,  
  width: 100,  
  color: Colors.red,
```



```
)
```

Example 1: Simple Container

```
Container(  
  height: 100,  
  width: 100,  
  color: Colors.green,  
);
```

Example 2: Padding & Margin

```
Container(  
  margin: EdgeInsets.all(10),  
  padding: EdgeInsets.all(15),  
  color: Colors.blue,  
  child: Text("Box"),  
);
```

Example 3: Decoration

```
Container(  
  height: 100,  
  width: 100,  
  decoration: BoxDecoration(  
    color: Colors.orange,  
    borderRadius: BorderRadius.circular(10),  
  ),  
);
```

Example 4: Container with Size, Color & Padding

```
Container(  
  width: 150,  
  height: 100,  
  padding: EdgeInsets.all(16),  
  color: Colors.blue,  
  child: Text(  
    'Blue Box',  
    style: TextStyle(color: Colors.white),  
  ),  
)
```

Example 5: Container with Margin, Border & Rounded Corners

```
Container(  
  margin: EdgeInsets.all(20),  
  padding: EdgeInsets.all(14),  
  decoration: BoxDecoration(  
    color: Colors.green,  
    border: Border.all(width: 3, color: Colors.black),  
    borderRadius: BorderRadius.circular(12),  
  ),  
  child: Text('Styled Container'),  
);
```

)

Simple Flutter Code

You can achieve that with the `borderRadius` property inside `BoxDecoration`:

```
Scaffold(  
  body: Center(  
    child: Container(  
      height: 200,  
      width: 300,  
      decoration: BoxDecoration(  
        color: Colors.blue,  
        borderRadius: BorderRadius.only(  
          topLeft: Radius.circular(40),  
          bottomLeft: Radius.circular(0),  
          topRight: Radius.circular(0),  
          bottomRight: Radius.circular(40),  
        ),  
        boxShadow: [  
          BoxShadow(  
            color: const Color.fromARGB(255, 212, 0, 0).withAlpha(50), // shadow color  
            blurRadius: 12, // softness  
            spreadRadius: 2, // size  
            offset: Offset(4, 6), // position (x, y)  
          ),  
        ],  
      ),  
    child: const Center(  
      child: Text(  
        "Custom Shape",  
        style: TextStyle(  
          color: Colors.white,  
          fontSize: 18,  
          fontWeight: FontWeight.w600,  
        ),  
      ),  
    ),  
  ),  
);
```

Basic Text

```
Text('Hello Flutter')
```

2 Styled Text

```
Text(  
  'Welcome',  
  style: TextStyle(  
    fontSize: 20,
```

```
fontWeight: FontWeight.bold,  
color: Colors.blue,  
),  
)
```

3 Text Alignment

```
Text(  
  'Center Text',  
  textAlign: TextAlign.center,  
)
```

4 Max Lines & Overflow

```
Text(  
  'This is a very long text that may overflow',  
  maxLines: 2,  
  overflow: TextOverflow.ellipsis,  
)
```

5 Custom Font (from pubspec.yaml)

```
Text(  
  'Custom Font Text',  
  style: TextStyle(  
    fontFamily: 'Poppins',  
    fontSize: 18,  
  )),  
)
```

6 Rich Text (Multiple Styles in One Line)

```
RichText(  
  text: TextSpan(  
    text: 'Hello ',  
    style: TextStyle(color: Colors.black, fontSize: 16),  
    children: [  
      TextSpan(  
        text: 'Flutter',  
        style: TextStyle(  
          color: Colors.blue,  
          fontWeight: FontWeight.bold,  
        )),  
    ],  
  ),  
)
```

7 Gradient Text

```
ShaderMask(  
  shaderCallback: (bounds) => LinearGradient(  
    colors: [Colors.red, Colors.orange],  
  ).createShader(bounds),  
  child: Text(  
    'Gradient Text',  
    style: TextStyle(  
      fontSize: 24,  
      color: Colors.white,  
      fontWeight: FontWeight.bold,  
    ),  
  ),  
)
```

8 Text with Shadow

```
Text(  
  'Shadow Text',  
  style: TextStyle(  
    fontSize: 22,  
    shadows: [  
      Shadow(  
        blurRadius: 5,  
        color: Colors.black45,  
        offset: Offset(2, 2),  
      ),  
    ],  
  ),  
)
```

9 Selectable Text

```
SelectableText('You can select this text')
```

Basic Syntax

```
Row(  
  children: [  
    Widget1(),  
    Widget2(),  
    Widget3(),  
  ],  
)
```

What is `Row` in Flutter?

`Row` is a **layout widget** that arranges its children **horizontally** (left → right).

- Think of it like a **horizontal line** where widgets sit next to each other.

□ Basic Syntax

```
Row(  
  children: [  
    Widget1(),  
    Widget2(),  
    Widget3(),  
  ],  
)
```

□ Example 1: Simple Row (Text Widgets)

This is the most basic use of Row.

```
Row(  
  children: const [  
    Text("Name"),  
    SizedBox(width: 10),  
    Text(":"),  
    SizedBox(width: 10),  
    Text("Dhanashree"),  
  ],  
)
```

□ Output

Name : Dhanashree in a single horizontal line.

□ Example 2: Row with `MainAxisAlignment`

This controls **horizontal spacing**.

```
Row(  
  mainAxisAlignment: MainAxisAlignment.spaceBetween,  
  children: const [  
    Icon(Icons.home),  
    Icon(Icons.search),  
    Icon(Icons.settings),  
  ],  
)
```

□ What happens?

- Icons are spaced **evenly across the screen**
- Very common in **app bars, footers, menus**

Other useful options:

- `MainAxisAlignment.start`
- `MainAxisAlignment.center`
- `MainAxisAlignment.end`
- `MainAxisAlignment.spaceAround`
- `MainAxisAlignment.spaceEvenly`

□ **Example 3: Row with Expanded (Very Important □)**

Used to **share available width**.

```
Row(  
  children: [  
    Expanded(  
      flex: 2,  
      child: Container(  
        height: 50,  
        color: Colors.blue,  
        child: const Center(child: Text("Left")),  
      ),  
    ),  
    Expanded(  
      flex: 1,  
      child: Container(  
        height: 50,  
        color: Colors.green,  
        child: const Center(child: Text("Right")),  
      ),  
    ),  
  ],  
)
```

□ **Explanation**

- Total space = 3 parts
- Blue gets **2 parts**
- Green gets **1 part**
- Perfect for **forms, dashboards, split layouts**

□ **Bonus: Row with CrossAxisAlignment**

Controls **vertical alignment**.

```
Row(  
  crossAxisAlignment: CrossAxisAlignment.center,  
  children: const [  
    Icon(Icons.person, size: 40),  
    SizedBox(width: 10),  
    Text("Profile", style: TextStyle(fontSize: 20)),  
  ],  
)
```

```
],  
)
```

☐ Common Mistake (Overflow Error)

☐ This causes overflow:

```
Row(  
  children: [  
    Text("Very very very long text"),  
    Text("More text"),  
  ],  
)
```

☐ Fix using Expanded:

```
Row(  
  children: const [  
    Expanded(child: Text("Very very very long text")),  
    Text("More"),  
  ],  
)
```

What is `Column` in Flutter?

`Column` is a **layout widget** that arranges its **children vertically** (top → bottom).

☐ It's commonly used for:

- Forms
- Profile screens
- Login pages
- Cards & sections

☐ Basic Syntax

```
Column(  
  children: [  
    Widget1(),  
    Widget2(),  
    Widget3(),  
  ],  
)
```

☐ Example 1: Simple Vertical Text Layout

Use case: Display texts one below another.

```
Column(  
  children: const [  
    Text(  
      'Flutter',  
      style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),  
    ),  
    Text('UI Toolkit'),  
    Text('By Google'),  
  ],  
)
```

□ **Output:**

```
Flutter  
UI Toolkit  
By Google
```

□ **Example 2: Column with Alignment & Spacing**

Use case: Center content vertically and horizontally.

```
Column(  
  mainAxisAlignment: MainAxisAlignment.center, // vertical  
  crossAxisAlignment: CrossAxisAlignment.center, // horizontal  
  children: const [  
    Icon(Icons.flutter_dash, size: 60, color: Colors.blue),  
    SizedBox(height: 10),  
    Text(  
      'Welcome to Flutter',  
      style: TextStyle(fontSize: 18),  
    ),  
  ],  
)
```

□ **Key Properties:**

- `mainAxisAlignment` → vertical alignment
 - `crossAxisAlignment` → horizontal alignment
 - `SizedBox` → spacing between widgets
-

□ **Example 3: Column Inside Container (Card-like UI)**

Use case: Profile / Info card.

```
Container(  
  padding: const EdgeInsets.all(16),  
  decoration: BoxDecoration(  
    color: Colors.white,  
    borderRadius: BorderRadius.circular(12),  
  ),  
)
```



```

boxShadow: [
  BoxShadow(
    color: Colors.black.withOpacity(0.1),
    blurRadius: 10,
    offset: const Offset(0, 4),
  ),
],
),
child: Column(
  crossAxisAlignment: CrossAxisAlignment.start,
  children: const [
    Text(
      'Dhanashree Wagh',
      style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
    ),
    SizedBox(height: 8),
    Text('Flutter Developer'),
    SizedBox(height: 4),
    Text('UI/UX Designer'),
  ],
),
)

```

- ❑ Very common in dashboards, lists & cards.

❑ Example 4: Column with Expanded Widgets

Use case: Divide vertical space equally.

```

Column(
  children: [
    Expanded(
      child: Container(color: Colors.red),
    ),
    Expanded(
      child: Container(color: Colors.green),
    ),
    Expanded(
      child: Container(color: Colors.blue),
    ),
  ],
)

```

- ❑ Each container takes **equal height**.

❑ Important Column Properties (Quick Table)

Property	Purpose
children	Widgets inside Column
mainAxisAlignment	Vertical alignment

Property	Purpose
crossAxisAlignment	Horizontal alignment
mainAxisSize	Min or Max height
Expanded	Flexible height

❑ Common Mistake

❑ Overflow error

```
Column(  
  children: [  
    // too many widgets without scroll  
  ],  
)
```

❑ Fix using SingleChildScrollView

```
SingleChildScrollView(  
  child: Column(  
    children: [],  
  ),  
)
```

```
import 'package:flutter/material.dart';
```

```
class RowColumnExample extends StatelessWidget {
```

```
  const RowColumnExample({super.key});
```

```
  @override
```

```
  Widget build(BuildContext context) {
```

```
    return Scaffold(  
      appBar: AppBar(  
        title: const Text("Row & Column Example"),  
      ),  
      body: Center(  
        child: Container(  
          width: 300,
```

```
padding: const EdgeInsets.all(16),

decoration: BoxDecoration(

  color: Colors.blue.shade50,

  borderRadius: BorderRadius.circular(16),

  boxShadow: [

    BoxShadow(

      color: Colors.black.withOpacity(0.15),

      blurRadius: 8,

      offset: const Offset(0, 4),

    ),

  ],

),

// COLUMN → vertical layout

child: Column(

  mainAxisAlignment: MainAxisAlignment.min,

  crossAxisAlignment: CrossAxisAlignment.start,

  children: [

    // ROW → horizontal layout

    Row(

      children: const [

        Icon(

          Icons.flutter_dash,

          color: Colors.blue,

          size: 30,

        ),
```

```
      SizedBox(width: 8),

      Text(
        "Flutter UI",
        style: TextStyle(
          fontSize: 20,
          fontWeight: FontWeight.bold,
        ),
      ),
    ],
  ),
```

```
const SizedBox(height: 8),
```

```
const Text(
  "Simple Row and Column layout",
  style: TextStyle(
    fontSize: 14,
    color: Colors.black54,
  ),
),
```

```
const SizedBox(height: 12),
```

```
// Another ROW inside COLUMN
```

```
Row(
  children: const [
    Icon(
```

```
Icons.star,  
  
color: Colors.orange,  
  
size: 20,  
  
),  
  
SizedBox(width: 6),  
  
Text(  
  
  "Rating: 4.8",  
  
  style: TextStyle(  
  
    fontSize: 16,  
  
  ),  
  
),  
  
],  
  
),  
  
],  
  
),  
  
),  
  
),  
  
),  
  
);  
  
}  
  
}
```